

SMARTBRIDGE / SKILL WALLET

Code-Layout, Readability and Reusability

Date 30 September 2026
Team ID SWTID-2026-6777
Project Name ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks 5 Marks

Code-Layout, Readability and Reusability:
This document evaluates the quality of the ComicCraft codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent Python indentation and formatted frontend code.
2	Proper File Structure	Files and folders are logically organized	Yes	Backend modules, templates, static files, configuration and export components are separated.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive names are used for stories, prompts, responses, layouts and configuration.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as generate_text, image generation, layout building and export handlers are clearly named.
5	Code Comments	Inline and block comments explain logic	Partial	Core code is readable; additional comments can be added for complex AI and generation logic.
6	Modular Design	Code is split into reusable functions/modules	Yes	Gemini Flash, Gemini Pro, image generation, layout building and export logic are separated.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Shared configuration and AI responsibilities are organized into reusable modules.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Validation and API error handling are included for invalid requests and generation failures.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	gemini_flash.py	Python / Google Gemini	Story generation, prompt processing and creative content generation	High
2	gemini_pro.py	Python / Google Gemini	Detailed story planning and advanced comic-content generation workflows	High
3	image_generator.py	Python / AI image generation	Comic character/scene image generation and panel visuals	High
4	layout_builder.py	Python / FastAPI	Comic panel layout and page composition workflow	High
5	exporters.py	Python / PDF export	Comic preview and final PDF export workflow	High

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Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear separation of application, AI modules, frontend assets and export components.
Readability	4	Descriptive names and straightforward request/response flows; some complex AI logic could use more comments.
Reusability	5	Task modules are separated and shared Gemini functionality avoids duplicated API setup.
Documentation / Comments	4	README and module boundaries document the project; additional inline comments can improve maintainability.
Overall Score	4.5 / 5	Strong modular structure with good readability and reusable components; minor documentation improvements remain.

Implementation Summary:

Area	ComicCraft Implementation
Backend	FastAPI REST API with separate AI, story, layout and export modules and request validation.
Frontend	HTML, CSS and JavaScript interface for entering story ideas, displaying generated comic content and previewing results.
AI Integration	Google Gemini models are integrated through dedicated Gemini modules for AI-powered comic story creation.
Image Generation	AI image generation is organized as a reusable component for comic characters, scenes and visual panels.
Testing	API and module tests can cover validation, generation workflows and mocked AI responses.
Maintainability	Configuration is environment-based and AI, layout and export responsibilities are separated into reusable modules.